

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech (EC) Theory Examination

May 2018

EC746/EC712 RF & Wireless Communication Systems

Date: 05.05.2018, Saturday

Time: 1:30 pm To 4:30 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following questions [07]

(a) Give the definitions of the following: [03]

- 1) Return loss,
- 2) Standing wave,
- 3) Skip zone

(b) A loss less line of characteristic impedance Z_0 is terminated in pure reactance of $-jZ_0$ value. VSWR is _____ [01]

A. 10 B. 2 C. 1 D. infinity

(c) The reflection coefficient on a line is $0.2 \angle 45^\circ$. The SWR is _____ [01]

A. 0.8 B. 1.1 C. 1.2 D. 1.5

(d) If Lossless line $Z_L < Z_0$, then [01]

- A. Reflection coefficient is zero
B. Reflection coefficient is ∞
C. Reflection coefficient is negative
D. Reflection coefficient is positive

(e) If $Z_0 = 100$ and $Z_L = 25$, reflection coefficient at load is [01]

A. 0.6 B. - 0.6 C. 0.4 D. - 0.4

Q-2 (a) Discuss various layers of ionosphere. Compare them with respect to their electrical parameters in tabulated fashion. [04]

Q-2 (b) A low transmission line having characteristics impedance 50Ω is terminated by load impedance $= 100 + 100j \Omega$. Using Smith chart, find [10]

- a) line impedance at $d = 0.5\lambda$ from load
- b) line impedance at $d = 0.1\lambda$ from load
- c) SWR
- d) distance of first $V_{\max}$ and $V_{\min}$ from load
- e) reflection coefficient at load (magnitude and angle)

OR

Q-2 (a) Explain Okumura model in detail. [07]

Q-2 (b) Design a high pass filter with pass band ripple magnitude less than 0.01dB. It must pass all frequencies over 100MHz and exhibit at least 5 dB attenuation at 25MHz. Load and source resistance is 75Ω . [07]

Q-3 (a) What do you mean by radio wave propagation? Explain Interference Mechanisms in detail. [07]

Q-3 (b) A cellular service provider decides to use a TDMA scheme that can tolerate an SIR of 16 dB in the worst case. Find the optimum value of cluster size N in case of (a) an omnidirectional antenna (b) 120° sectoring (c) 60° [07]

sectoring. What will be the advantage of sectoring? Which sectoring is better 120° or 60° ? Assume path loss exponent $n=4$

OR

- Q-3 (a)** Explain Hata model in detail. [07]
Q-3 (b) What is diffraction? Explain it with knife edge diffraction model. [07]

SECTION- II

- Q-4(a)** Define the following: [03]
1) Erlang
2) Cell dragging
3) MAHO
Q-4(b) Explain microcell zone concept [04]
Q-5 (a) What do you mean by fading? Explain factors influencing small scale fading. [07]
Q-5 (b) Explain different types of propagations. [07]

OR

- Q-5 (a)** Design $\lambda/2$ long microstrip resonator using 50 ohm line that is short circuited at its ends. The substrate thickness is 1.59mm with dielectric constant 2.08 and loss tangent 0.0004. [05]
Q-5 (b) Design an air filled circular cylindrical cavity that resonates at 5GHz in TE₀₁₁ mode. It should be made up of copper and its height should be equal to its diameter. [02]
Q-5 (c) Prove that for hexagonal geometry, the co-channel reuse ratio is given by $Q=N^2$, where $N=i^2+j^2$. [07]
Q-6(a) Explain artificial intelligent network characteristics. [07]
Q-6 (b) Design a three element maximally flat low pass filter with its cutoff frequency at 1GHz. It is to be used between a 50 Ω and a generator with its internal impedance at 50 Ω . Assume $Z_h=150\Omega$ and $Z_m=30\Omega$ ($g_0=g_4=1, g_1=1, g_2=2, g_3=1$) [07]

OR

- Q-6 (a)** Compare channel sharing and channel borrowing. [07]
Q-6 (b) What is handoff? Illustrate the handoff scenario at cell boundary. [07]
